

DBMS – ORDER AND ORDER_ITEM MANAGEMENT

HUFT – E-Commerce Order Management Database System

1.DATABASE DESIGN

1.1 ORDER TABLE

Attribute	Description
Order_ID	Unique product ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

1.2 ORDER_ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2.RELATIONSHIP



3.TABLE CREATION

3.1 ORDER TABLE

```
SQL> CREATE TABLE HuftOrder (  
2     Order_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER,  
4     Order_Date DATE NOT NULL,  
5     Total_Amount NUMBER(10,2),  
6     Order_Status VARCHAR2(20)  
7 );  
  
Table created.
```

3.2 ORDER_ITEM TABLE

```
SQL> CREATE TABLE HuftOrder_Item (  
2     Order_Item_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER REFERENCES HuftOrder(Order_ID),  
4     Product_ID NUMBER REFERENCES HuftProduct(Product_ID),  
5     Quantity NUMBER NOT NULL,  
6     Price NUMBER(10,2),  
7     Total_Amount NUMBER(10,2)  
8 );  
  
Table created.
```

4.ORDER TABLE DATA INSERTION

```
SQL> INSERT INTO HuftOrder VALUES  
2 (1001, 1, TO_DATE('18-09-2026','DD-MM-YYYY'), 1750.00, 'Delivered');  
  
1 row created.  
  
SQL> INSERT INTO HuftOrder VALUES  
2 (1002, 2, TO_DATE('18-09-2026','DD-MM-YYYY'), 1849.00, 'Shipped');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO HuftOrder VALUES  
2 (1003, 3, TO_DATE('19-09-2026','DD-MM-YYYY'), 1099.00, 'Processing');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO HuftOrder VALUES  
2 (1004, 4, TO_DATE('19-09-2026','DD-MM-YYYY'), 1273.00, 'Delivered');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO HuftOrder VALUES  
2 (1005, 5, TO_DATE('20-09-2026','DD-MM-YYYY'), 1075.00, 'Shipped');  
  
1 row created.
```

```

SQL> INSERT INTO HuftOrder VALUES
  2 (1006, 6, TO_DATE('20-09-2026','DD-MM-YYYY'), 650.00, 'Processing');

1 row created.

SQL>
SQL> INSERT INTO HuftOrder VALUES
  2 (1007, 7, TO_DATE('20-09-2026','DD-MM-YYYY'), 1197.00, 'Delivered');

1 row created.

SQL>
SQL> INSERT INTO HuftOrder VALUES
  2 (1008, 8, TO_DATE('21-09-2026','DD-MM-YYYY'), 1498.00, 'Processing');

1 row created.

SQL>
SQL> INSERT INTO HuftOrder VALUES
  2 (1009, 9, TO_DATE('21-09-2026','DD-MM-YYYY'), 998.00, 'Shipped');

1 row created.

SQL>
SQL> INSERT INTO HuftOrder VALUES
  2 (1010, 10, TO_DATE('21-09-2026','DD-MM-YYYY'), 900.00, 'Processing');

1 row created.

```

4.1 DISPLAY ORDER TABLE DATA

```

SQL> SELECT * FROM HuftOrder;

ORDER_ID  CUSTOMER_ID  ORDER_DAT  TOTAL_AMOUNT  ORDER_STATUS
-----
1001      1 18-SEP-26    1750 Delivered
1002      2 18-SEP-26    1849 Shipped
1003      3 19-SEP-26    1099 Processing
1004      4 19-SEP-26    1273 Delivered
1005      5 20-SEP-26    1075 Shipped
1006      6 20-SEP-26     650 Processing
1007      7 20-SEP-26    1197 Delivered
1008      8 21-SEP-26    1498 Processing
1009      9 21-SEP-26     998 Shipped
1010     10 21-SEP-26     900 Processing

10 rows selected.

```

5.ORDER_ITEM TABLE DATA INSERTION

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (1, 1001, 101, 2, 650.00, 1300.00);
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (2, 1001, 103, 1, 450.00, 450.00);
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (3, 1002, 102, 1, 749.00, 749.00);
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (4, 1002, 106, 2, 550.00, 1100.00);
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (5, 1003, 104, 2, 350.00, 700.00);
```

1 row created.

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (6, 1003, 107, 1, 399.00, 399.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (7, 1004, 108, 2, 275.00, 550.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (8, 1004, 109, 1, 499.00, 499.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (9, 1005, 110, 1, 425.00, 425.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (10, 1005, 112, 2, 325.00, 650.00);
```

```
1 row created.
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (11, 1006, 101, 1, 650.00, 650.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (12, 1007, 107, 3, 399.00, 1197.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (13, 1008, 102, 2, 749.00, 1498.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (14, 1009, 109, 2, 499.00, 998.00);
```

```
1 row created.
```

```
SQL>
```

```
SQL> INSERT INTO HuftOrder_Item VALUES  
2 (15, 1010, 103, 2, 450.00, 900.00);
```

```
1 row created.
```

5.1 DISPLAY ORDER_ITEM TABLE DATA

```
SQL> SELECT * FROM HuftOrder_Item;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	TOTAL_AMOUNT
1	1001	101	2	650	1300
2	1001	103	1	450	450
3	1002	102	1	749	749
4	1002	106	2	550	1100
5	1003	104	2	350	700
6	1003	107	1	399	399
7	1004	108	2	275	550
8	1004	109	1	499	499
9	1005	110	1	425	425
10	1005	112	2	325	650
11	1006	101	1	650	650

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	TOTAL_AMOUNT
12	1007	107	3	399	1197
13	1008	102	2	749	1498
14	1009	109	2	499	998
15	1010	103	2	450	900

15 rows selected.

6. ORDER MODIFICATION OPERATIONS

6.1 MODIFY ORDER AMOUNT

```
SQL> UPDATE HuftOrder
  2 SET Total_Amount = 1800.00
  3 WHERE Order_ID = 1001;
```

1 row updated.

```
SQL> SELECT *
  2 FROM HuftOrder
  3 WHERE Order_ID = 1001;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
1001	1	18-SEP-26	1800	Delivered

6.2 MODIFY ORDER DATE

```
SQL> UPDATE HuftOrder
  2  SET Order_Date = TO_DATE('22-09-2026', 'DD-MM-YYYY')
  3  WHERE Order_ID = 1002;
```

1 row updated.

```
SQL> SELECT *
  2  FROM HuftOrder
  3  WHERE Order_ID = 1002;
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
1002	2	22-SEP-26	1849	Shipped

7. ORDER_ITEM MODIFICATION

```
SQL> UPDATE HuftOrder_Item
  2  SET Quantity = 3,
  3      Total_Amount = 1050.00
  4  WHERE Order_Item_ID = 5;
```

1 row updated.

```
SQL> SELECT *
  2  FROM HuftOrder_Item
  3  WHERE Order_Item_ID = 5;
```

ORDER_ITEM_ID	ORDER_ID	PRODUCT_ID	QUANTITY	PRICE	TOTAL_AMOUNT
5	1003	104	3	350	1050

```
SQL> UPDATE HuftOrder
  2  SET Total_Amount = 1449.00
  3  WHERE Order_ID = 1003;
```

1 row updated.

8. CUSTOMER ORDER HISTORY REPORT

```
SQL> SELECT
  2     c.CID AS Customer_ID,
  3     c.CNAME AS Customer_Name,
  4     o.Order_ID,
  5     o.Order_Date,
  6     oi.Product_ID,
  7     p.Product_Name,
  8     oi.Quantity,
  9     oi.Price,
 10     oi.Total_Amount
 11 FROM CUST3 c
 12 JOIN HuftOrder o
 13 ON c.CID = o.Customer_ID
 14 JOIN HuftOrder_Item oi
 15 ON o.Order_ID = oi.Order_ID
 16 JOIN HuftProduct p
 17 ON oi.Product_ID = p.Product_ID
 18 ORDER BY c.CID, o.Order_Date;
```

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

1	Abi	1001	18-SEP-26	101
Chicken Dog Food				
2	650	1300		

1	Abi	1001	18-SEP-26	103
Leather Dog Collar				
1	450	450		

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

2	Anu	1002	22-SEP-26	106
Cat Litter				
2	550	1100		

2	Anu	1002	22-SEP-26	102
Grain Free Cat Food				

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

1	749	749		
3	Ravi	1003	19-SEP-26	104
Pet Grooming Shampoo				
3	350	1050		
3	Ravi	1003	19-SEP-26	107
CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

Dental Chew Sticks				
1	399	399		
4	Karthik	1004	19-SEP-26	108
Pet Feeding Bowl				
2	275	550		

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

4	Karthik	1004	19-SEP-26	109
Puppy Training Pads				
1	499	499		
5	Priya	1005	20-SEP-26	112
Cat Grooming Brush				
2	325	650		
CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

5	Priya	1005	20-SEP-26	110
Pet Conditioner				
1	425	425		
6	Divya	1006	20-SEP-26	101
Chicken Dog Food				

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

1	650	650		
7 Arun		1007	20-SEP-26	107
Dental Chew Sticks				
3	399	1197		
8 Meena		1008	21-SEP-26	102

CUSTOMER_ID	CUSTOMER_NAME	ORDER_ID	ORDER_DAT	PRODUCT_ID

PRODUCT_NAME				

QUANTITY	PRICE	TOTAL_AMOUNT		

Grain Free Cat Food				
2	749	1498		
10 Nisha		1010	21-SEP-26	103
Leather Dog Collar				
2	450	900		

14 rows selected.

9. CUSTOMER-WISE ORDER SUMMARY

```
SQL> SELECT
  2     c.CID AS Customer_ID,
  3     c.CNAME AS Customer_Name,
  4     COUNT(o.Order_ID) AS Total_Orders,
  5     SUM(o.Total_Amount) AS Total_Amount
  6 FROM CUST3 c
  7 JOIN HuftOrder o
  8 ON c.CID = o.Customer_ID
  9 GROUP BY c.CID, c.CNAME
 10 ORDER BY c.CID;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS	TOTAL_AMOUNT
1	Abi	1	1800
2	Anu	1	1849
3	Ravi	1	1449
4	Karthik	1	1273
5	Priya	1	1075
6	Divya	1	650
7	Arun	1	1197
8	Meena	1	1498
10	Nisha	1	900

9 rows selected.